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# Web-Based Five Senses Input Simulation Twenty Years Later – in the Era of LLMs

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**Abstract:** Perception plays a crucial role in human learning and influences our knowledge acquisition. On the other hand, sensory capabilities of machines are still limited, especially when it comes to touch, taste and smell, which limits experiments with embodied agents. On the other hand, current large language models show various capabilities which might be useful to augment perception. Following our two previous papers discussing the possibility of retrieving hints for future physical agents from text, in this paper we report findings from our preliminary experiments with commercial large language models, namely GPT-3.5 and GPT-4, by feeding them with real world-describing sentences (in English and Japanese) which are altered to represent physical changes. The experimental results demonstrate some improvement, while also revealing significant challenges associated with the approach, data, and evaluation methods.

## 1 Introduction

Recent success of large language models (LLMs) has inspired research not only in the field of natural language processing. For example, [Driess 23] propose PaLM-E, an embodied language model which directly incorporates real-world continuous sensor modalities into language models. Establishing the link between words and percepts allows to control different robots in simulation and in the real world. However, combining sensory input with text concentrates on vision, ignoring other senses as sound, touch, smell or taste, which also play a role in human cognitive processes. This limitation is due to the lack or cost of hardware which make experiments almost impossible, especially for less-funded research units. To tackle this problem, in the past we proposed [Rzepka 03, Rzepka 13] to utilize vast Internet resources to retrieve common sensory information and meaning of situations when something uncommon happens to an object. For example, if common colors of an apple are red, green or yellow, and a robot discovers a brown apple, it can hypothesize that the fruit is rotten and its other features like smell and softness also change. This could lead to the changes of the robot's behavior – such apple should not be given to a child, desirably should be disposed in a odor-resistant trash bag, and strength of the robotic fingers should be adjusted. In the real world, sudden and unpredictable changes of an environment or agents within it, are practically unavoidable. In the previous experiments we showed that retrieving such knowledge from the text was limited due to the high precision but low recall. In the case of large language models, search failures are rare, therefore we decided to test their capability to recognize changes in the context. In this short paper we report the results and describe problems we discovered.

## 2 Related Work

Research on importance of perception in grounding language has focused on the dominant senses of sight and hearing while relatively fewer studies have tackled other modalities in sensorimotor activation which is critical to language processing. [Speed 20] review works in cognitive science on simulation within touch, taste, and smell and conclude that perceptual activation from lesser studied modalities weakly contribute to sensorimotor simulation and grounding meaning in language. However, as the author mention, further, more refined research is needed because it is not clear whether brain activations used in human experiments sufficiently reflect mental simulation or mental imagery. When it comes to five senses and large language models, the only research we found is done by [Chakrabarty 23] who only mention sensory input in a prompt for increasing narrative skills of some current models. Recently, research on qualia within the

context of Large Language Models (LLMs) has been published [Ding 23], but we are more interested in commonsense sensory information rather than subjective and personal experience of such knowledge.

Research on sensing and language limited to vision has become popular in recent years. Except PaLM-E mentioned in the previous section [Driess 23], there are multiple examples of implementing multimodal input in robotics, including object manipulation, recognition and affordances [Ahn 22]. Although it has become popular to test cognitive abilities large language models, they are usually about mental, not physical realm [Binz 23]. To the authors' best knowledge, our experiments are the first attempt to test all multimodal sensory information contained in LLMs.

# 3 Experimental Data

In order to observe if a large language model can recognize possible changes of perception (five basic senses) in text, we first prepare a small set of sentences describing the world and alternate them to indicate change which could be perceived by one of the senses. We randomly select image descriptions from *flickr* dataset[1] to assure that input sentences describe physical world. If a sentence contains direct description of an object (e.g. "white shirt"), we discard it and choose another one until we reach five in total. In the next step we manually alter the sentences to convey a new sensory aspect of a chosen object, creating the test dataset of 30 sentences shown in Table 1. Then we create a prompt asking a model to select the most natural phrase (usually an adjective) from a set of sense-word lists. The set is borrowed from [Rzepka 03, Rzepka 13], but because the original lists of phrases describing senses are rather limited, we use Bard[2] to add missing sensory words for every of five categories. After removing (most) multi-word, subjective (e.g. "nice") and metaphorical phrases, the final extended set consist of 56 words for visual, 40 for auditory, 40 for haptic, 16 for gustatory and 14 for olfactory senses. Phrases indicating that an object has no perceivable sense ("no smell", "no taste", etc.) have been manually added by the first author.

[1] `https://www.kaggle.com/datasets/hsankesara/flickr-image-dataset`

[2] `http://bard.google.com/` We chose this model heuristically by comparing outputs of various models and it generated the longest list with almost no repetitions. "Enlarge the list" was the prompt.

# 4 Experiments and Results

For the experiments use two recent models of Generative Pre-trained Transformer (GPT), namely ChatGPT[3] and GPT-4[4], to select most natural phrases describing an object from the corresponding lists. The temperature is set to 0 and the prompts we utilize have been formed as follows:

> **statement**: input sentence (see Table 1)
> **system_role**: "You are a robot with five senses perception capabilities, observing a situation."
> **question**: "Choose the best one word describing the SENSE[5] of the OBJECT from the following list (you can choose only one word, words from outside the list are not allowed):
> **answers**: list of words describing SENSE

After running both models, first and second authors performed a preliminary evaluation of model choices by assigning values from 1 to 5, where 1 indicates wrong (unnatural) description and 5 indicates correct (natural) choice of an object-describing sensory word. After averaging both scores, we found out that both models scored exactly the same (3.6), however there were four cases where GPT-4 generated a word which was not listed or refused answering, while GPT-3.5 did it only once. If these cases were excluded, GPT-4 score was 4.0 and GPT-3.5 average increased to 3.7. It must be noted that even if a model did not reply with a single choice but with a whole sentence, if a word from a list was in the sentence, we treated this word as the model's answer.

Because we struggled to find native English speakers for evaluation, we decided to translate the English dataset, sensory word lists and prompts to Japanese, rerun the experiment and compare its results to human choices. Instead of scoring the naturalness of models' choices, we decided to ask seven graduate students majoring in computer science to select the most appropriate sensory word from a list for each sentence. This decision was made to directly compare the outputs with human preferences. Because human evaluators completely agreed only in a single

[3] snapshot *gpt3.5-turbo-0613*

[4] snapshot *gpt4-0613*

[5] We adjusted the questions to sound naturally in English and avoided sense words in Visual examples to avoid suggesting a color or a shape as the answer.

Table 1: Original captions and their variations used for experiments

| Original Sentence | Altered Sentence (OBJECT marked with **bold** fonts) | SENSE |
|---|---|---|
| A young man on a train holds an **origami crane** | Man on a train pours water on an **origami crane** | Vision |
| | Man on a train crumpled an **origami crane** | Vision |
| | Many men on a train hold an **origami crane** together | Vision |
| | An old man on a train holds an **origami crane** | Vision |
| | A young man on a bus holds an **origami crane** | Vision |
| A group of four women are performing music on their **instruments** | A group of four women stopped performing music on their **instruments** | Sound |
| | A group of four women cutting their **instruments** into pieces | Sound |
| | A group of four women packing their **instruments** | Sound |
| | A group of four women dragging their heavy **instruments** across the river | Sound |
| | A group of four women dragging their heavy **instruments** across the snow | Sound |
| A man is sitting on a chair holding a large stuffed **animal** | A man is sitting on a chair holding a straw **animal** | Touch |
| | A man is sitting on a chair holding a real **animal** | Touch |
| | A man is standing and holding a large stuffed **animal** | Touch |
| | A man is sitting on a chair stroking a large stuffed **animal** | Touch |
| | A woman is sitting on a chair holding a large stuffed **animal** | Touch |
| An older woman is working behind a counter at a **restaurant** | An older woman is preparing fried rice behind a counter at a **restaurant** | Smell |
| | An older woman is behaving fishy behind a counter at a **restaurant** | Smell |
| | A young woman spilled beer behind a counter at a **restaurant** | Smell |
| | An older woman smells a rat behind a counter at a **restaurant** | Smell |
| | An older woman is cutting meat behind a counter at a at a **restaurant** | Smell |
| Girl in black jacket sifting powdered sugar over a chocolate **cake** | Girl in black jacket sifting powdered cocoa over a chocolate **cake** | Taste |
| | Girl in black jacket sifting powdered salt over a chocolate **cake** | Taste |
| | Girl in black jacket sifting powdered sugar over a white **cake** | Taste |
| | Girl in black jacket smearing cream on chocolate **cake** | Taste |
| | Boy in black jacket sifting powdered sugar over a chocolate **cake** | Taste |

Table 2: Accuracy rates depending on how many identical human-selected labels are treated as the correct answer

| | GPT-3.5 | GPT-4 |
|---|---|---|
| Loose (one) | 0.70 (21/30) | 0.63 (19/30) |
| Semi-strict (two) | 0.46 (14/30) | 0.36 (11/30) |
| Strict (three) | 0.33 (10/30) | 0.30 (9/30) |

case (sentence "A group of four women are performing music on their instruments" has been labeled as *melodious* by all seven annotators), we decided to calculate the correctness of automatic labeling on three levels of strictness: when the output word agreed at least with one (loose), with two (semi-strict) and three (strict) human annotators. The results are given in Table , but it must be noted that only 70% (21/30) of sentences had three or more identical gold labels, hence the Strict level should be considered as overly stringent.

# 5 Analysis and Discussion

When compared to the results on the English sentences, GPT-3.5 refused choosing or chose out-of-the-list words more often (2 times) and GPT-4 less often (3 times). Nevertheless, GPT-3.5 has yielded better results than GPT-4. However, three primary issues must be addressed in the future to establish any definitive conclusions.

Firstly, the set of sensory primitives has to be refined and probably restructured. For example, "touch" category could be divided into two subcategories, namely "texture" and "temperature", also "vision" could be separated into "shape" and "color". By doing so, we could simplify the human annotation as objects can be warm and fluffy at the same time. Also we found problems with automatic translations of word lists. For example "臭い" and "悪臭の" (*smelly*) are synonyms and could be merged, which could improve the agreement between annotators. Some English words were translated to identical Japanese words and were discarded as duplicates (Japanese lists have 52 words for visual, 36 for auditory, 33 for haptic, 16 for gustatory and 13 for olfactory senses). Another problem was the sentence translation task. A native Japanese speaker was asked to translate English sentences, and they made changes to ensure naturalness. For instance, idioms like "smell a rat" were replaced, as literal translations could be incomprehensible.

Secondly, there is an issue of how many words should be allowed to be selected by both annotators and the models. We have limited a model's choice to one in

order to observe if an LLM is able to select a single most characteristic sensory description. This partially simplifies the evaluation, but usually many words can describe one object even within a single sense category, hence this approach is unnatural.

Thirdly, we have noticed that models sometimes refuse to select a word by explaining that a robot does not possess five senses, therefore it is not able to choose the best-describing phrase. It is necessary to experiment with different *system roles* which are added to the prompt.

The annotation and the model outputs have showed the difficulty of deciding which physical features are representative in a given context. First associations we make are often rooted in our experiences and might be difficult to separate from subjective qualia which leaves an open question how machines could ground language. For instance, if an annotator is a vegan, her stronger association for the smell of meat being cut will most probably be different than a person who loves cooking steaks. This problem could be partly solved by limiting word lists to very primitive, basic phrases, but such limitation of a sensory reservoir can decrease the role of senses in learning or reasoning.

## 6 Conclusions and Future Work

In this short paper we presented the results of a simple experiment to observe if large language models choose the same words describing five senses as humans when faced with textual input about the real-world objects. The conclusion is that they cannot in most cases, but it is also difficult for human annotators. However, to build a text-based sensory simulator, various changes to the data and the experimental setup must be considered. In the next step, we use the findings of this research to change our approach and instead of concentrating on matching most relevant sensory words, we will perform experiments eliminating obviously incorrect associations. We are also preparing data in other languages to measure the generality of the words. Multilingual approach might be also useful to determine the set of primitives describing human senses.

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